Synthesis and Characterization of CdSe/ZnS Quantum Dots and Silver Nanoparticles

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Goal

Synthesis of CdSe/ZnS-based red quantum dots (QDs) and silver nanoparticles (AgNPs) for their use in Förster resonance energy transfer (FRET) quenching.

Introduction

- FRET is a phenomenon in which energy is transferred from an excited fluorophore donor to a non-fluorescent acceptor quencher without emission of visible light.
  - This means fluorescence can be quenched *via* the FRET mechanism, which is relevant to diagnostics and the study of protein-protein interactions, as depicted in 1.1.
- Quantum dots (QDs) are small fluorescent semiconductor nanocrystal (Ref. 4)
  - These have highly size-dependent optical properties, very high photostability, long fluorescent lifetimes (10 – 40 ns), and relatively high quantum yields (~30 to 90%) (Ref. 1, Ref. 4).
    - CdSe/ZnS core-shell QDs, as depicted in Fig. 1.2, were chosen for the relatively low cost of synthesis, high quantum yield, and low photobleaching.
    - The ZnS shell helps to improve quantum yield and reduce degradation of the CdSe core.
- Silver nanoparticles (AgNPs) are very effective fluorescence quenchers for QDs.
- We synthesized both CdSe/ZnS core-shell QDs and Ag nanoparticles successfully, performed spectroscopy of both, and tested quenching on water-solubilized commercial streptavidin-coated CdSe/ZnS QDs (Fisher Scientific) due to time constraints.

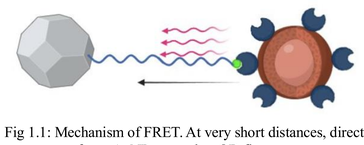


Fig 1.1: Mechanism of FRET. At very short distances, direct energy transfer to AgNPs quenches QD fluorescence

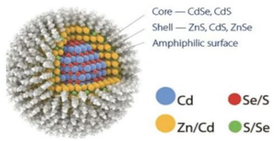


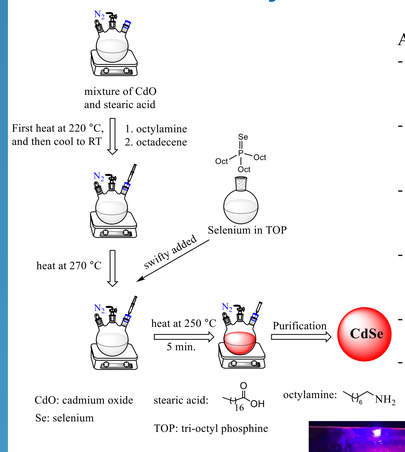
Fig 1.2: CdSe/ZnS core-shell QD structure

Synthesis of CdSe

Synthetic Procedure

As depicted in Scheme 2.1

- A suspension of CdO (0.2 mmol) and stearic acid (0.8 mmol) in a 25-mL three neck flask was heated at 220°C under N_2.
- After cooling to RT, 4.9 g octadecylamine (ODA) and 8 ml octadecene (ODE) were added into the flask, and reheated to 270°C.
- At this temperature, a Se-solution made by dissolving Se (2 mmol) in tri-n-octylphosphine (TOP, 2 mL) was injected.
- The growth temperature was then reduced to 250°C for 5 min and the reaction mixture was cooled to room temp.
- A mixture of ethanol and chloroform was used to purify the nanocrystals.
- TEM and fluorescence spectroscopy were used to characterized. Results in Fig 2.5, Fig 2.6



Scheme 2.1: Synthesis procedure for CdSe QDs by hot injection of TOP-Se



Fig 2.2: Reaction mixture under visible light (left image), and under UV (right image).

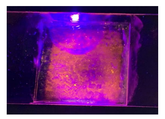


Fig 2.3: CdSe QD fluorescence, on microscope slide

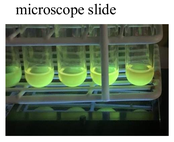


Fig 2.4: CdSe QD synthesis without ODA

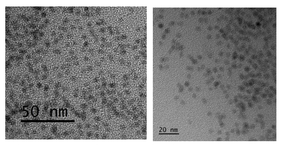


Fig 2.5: TEM images of CdSe QDs. 5.4 nm on average (ImageJ). 50 nm and 20 nm resolution

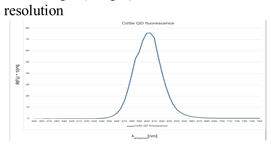
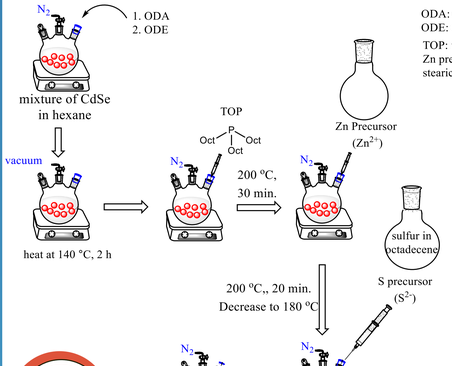


Fig 2.6: CdSe QD fluorescence in hexane. λ_{ex}: 400 nm

Synthesis of CdSe/ZnS Core-Shell QDs



ODA: Octadecyl amine
ODE: Octadecene
TOP: tri-octyl phosphine
Zn precursor: ZnO, and stearic acid in octadecene

Scheme 3.1: Synthesis of CdSe/ZnS shell based QD.

Synthetic Procedure (Scheme 3.1)

- Red CdSe QDs (5.4 nm in diameter, 7.7×10^{-5} mmol) dissolved in hexane (2.5 mL) were mixed with ODA (1.46 g), and ODE (4 mL) in a 25-mL three-neck flask.
- The hexane was removed by mechanical pump at RT for 30 min., followed by removing residual air at 100°C for another 10 min. System was switched to N_2 and heated to 140°C for 2 h.
- TOP (0.73 mL) was injected as an activator and the reaction mixture was further maintained at 200°C for 30 min.
- After the activation, the zinc oleate Zn-precursor solution (0.61 mL) was synthesized by dissolving ZnO (2 mmol) in oleic acid (15 mmol) and ODE (15 ml) at 290°C,
- Zn-precursor was injected (as shown in Scheme 3.1) and maintained at maintained at 200°C for 20 min. The temperature was then decreased to 180°C.
- At the same time, sulfur precursor (S-ODE, 0.1 mol/L) was prepared by dissolving sulfur in ODE at 130°C.
- The S-ODE (0.61 mL) was injected, and temperature of reaction mixture was increased to 220°C for 60 minutes to allow in-situ growth of first ZnS monolayer, and then decreased to 140°C.
- For the second ZnS layer, TOP (0.73 ml) was injected, mixture was heated to 180°C and held for 30 minutes, then Zn-precursor (0.84 ml) and S-ODE (0.84 ml) were injected, at 10 minutes intervals, then dropped to 140°C.
- TEM and fluorescence spectroscopy were used to characterized. Results in Fig 3.4, Fig 3.5



Fig 3.2: Comparison of CdSe QDs and CdSe/ZnS core/shell QD

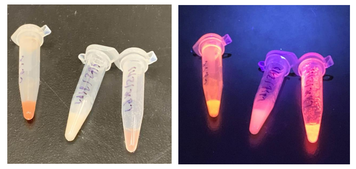


Fig 3.3: Comparison of CdSe and CdSe/ZnS QDs. Left CdSe, center CdSe/ZnS(crude), right CdSe/ZnS(purified), under visible and UV light.

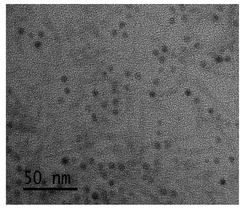


Fig 3.4: TEM of CdSe/ZnS QD. Average size was 6.87 nm. Image acquired at 50 nm

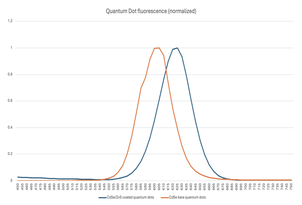


Fig 3.5: Normalized fluorescence spectra of CdSe and CdSe/ZnS QDs. λ_{ex} = 400 nm

Reductive Synthesis of Ag nanoparticles (NPs)



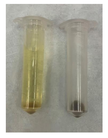
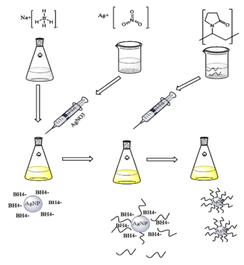


Fig 4.1: Samples of silver nanoparticles, before and after centrifugation

Scheme 4.1 Reductive synthesis with $NaBH_4$ and $AgNO_3$



Synthetic Procedure

- 10 ml 1.0 mM $AgNO_3$ in H_2O was added dropwise to 30 ml ice cold 2.0 mM $NaBH_4$ while stirred (shown in Scheme 4.1)
- 1 ml 0.3 % PVP40 solution (by weight) was added to coat the AgNPs to suppress aggregation.

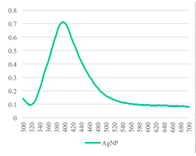
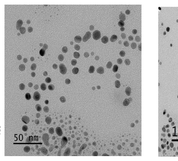


Fig 4.2: AgNP absorbance spectrum in H_2O



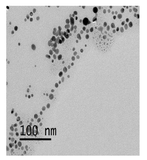


Fig 4.3: AgNP under TEM. Average size is 10.6 nm. Image acquired at 50 nm and 100 nm resolution

Quenching of CdSe/ZnS QD with Ag NP

- QD stock solution was made by mixing 1 mL PBS (pH = 7) and 1 µL quantum dots, then mixing 600 µL of that solution with 600 µL of ph7 PBS.
  - These were Invitrogen ITK 605 Nano quantum dots, which have already been water solubilized via polymer and streptavidin coating.
- 100 µl of QDs, plus the specified amount of 1:1 AgNP solution: PBS (pH = 7), were mixed together in 96-well plate, as seen in Fig 5.1
- 605 nm fluorescence was measured over the course of 1 hour (Fig 5.2

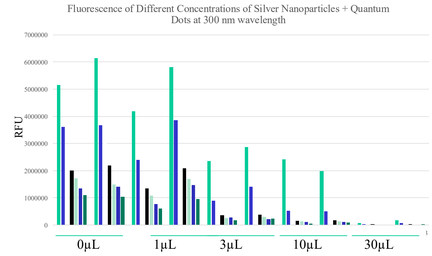


Fig 5.2: Fluorescence quenching data from spectrophotometry. Note significant quenching at 3 microliters of AgNP

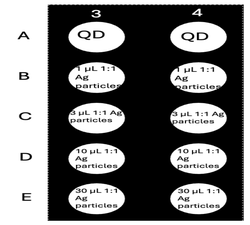


Fig 5.1: Wells used for quenching experiments, and silver nanoparticle amount added

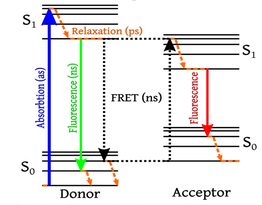


Fig 5.3: The mechanism of FRET. A fluorophore is energized, and virtual photon transfers energy to quencher or acceptor.

Conclusion

- We successfully synthesized red quantum dots using cadmium selenide (Fig 2.3) and successfully coated them with zinc sulfide to improve quantum yield and stability (Fig 3.3).
  - TEM images confirmed that the particles increased by 1.4 nanometers in diameter, suggesting at least 1 layer of ZnS was deposited (Ref. 1).
- We also successfully synthesized silver nanoparticles (Fig 4.2)
  - Quenching of water-solubilized CdSe/ZnS quantum dots was confirmed via spectrophotometry.
- We have demonstrated useable labels (CdSe/ZnS quantum dots and AgNP) for FRET/FRAP (Fig 5.2)

Acknowledgement

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Future Directions

- Measure quantum yield of QDs
- Water-solubilize and functionalize the QDs
  - Test quenching with our own QDs
  - Bioconjugate the QDs to streptavidin by carboxyl + NHS/EDC, for future use
- Test ways to use this quenching behavior to test for botulinum neurotoxin (as depicted in Fig 1.1)

References

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